

Principles of Economics

Problem Set 10

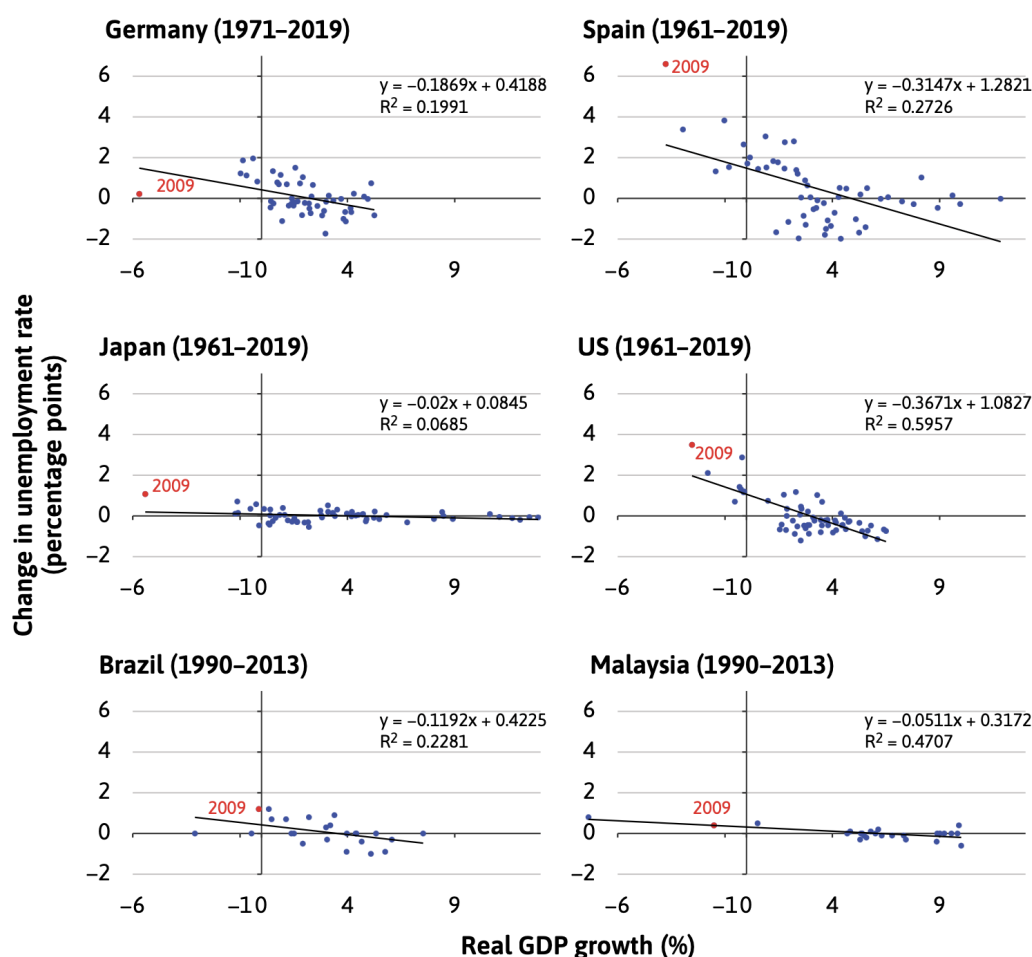
Universidad Carlos III de Madrid | Department of Economics

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1. WARMING-UP QUESTIONS

- Look at the regression lines (the lines of best fit) in Figure 13.5. What prediction does the regression line show for unemployment when the economy is not growing? Are the results the same for all the countries?

Figura 13.5



- Assume that the population in the economy is growing. Can you use this assumption to provide an explanation for your results in question 1?
- Consider a credit-constrained household type and a consumption smoothing household type.

- a. For each household type, use a figure with time on the horizontal axis and income and consumption on the vertical axis to explain the relationship between the change in income and the change in consumption when income returns to normal after an unexpected temporary decline.
- b. Based on this analysis, explain the predicted relationship between temporary changes in income and consumption for an economy with a mixture of the two household types

4. Data exercise.

- a. Use the data from FRED (<https://fred.stlouisfed.org>) to construct charts for real GDP growth, the unemployment rate, and the inflation rate for the US. Select the period from 1960 until the most recent year available. In addition, download the data for the US GDP deflator (search for GDPDEF). Use the data you downloaded to answer the following questions (remember that the CPI is calculated from the price of goods consumed in the home country, while the GDP deflator is calculated from the price of the goods produced in the home country):
- b. Observe the difference in the evolution of the series for the CPI and the GDP deflator that takes place in 1974–75 and 1979–1982. What could explain this pattern? (Hint: think about the likely impact of an oil crisis on the price of imported goods and, in particular, on your own transport and fuel bills.)
- c. What do you notice about the evolution of unemployment and inflation in the early 1980s?
- d. Now construct the same charts for your own country. Write a brief report on the evolution of inflation, unemployment, and the real GDP growth rate over the same period.

2. MULTIPLE CHOICE QUESTIONS

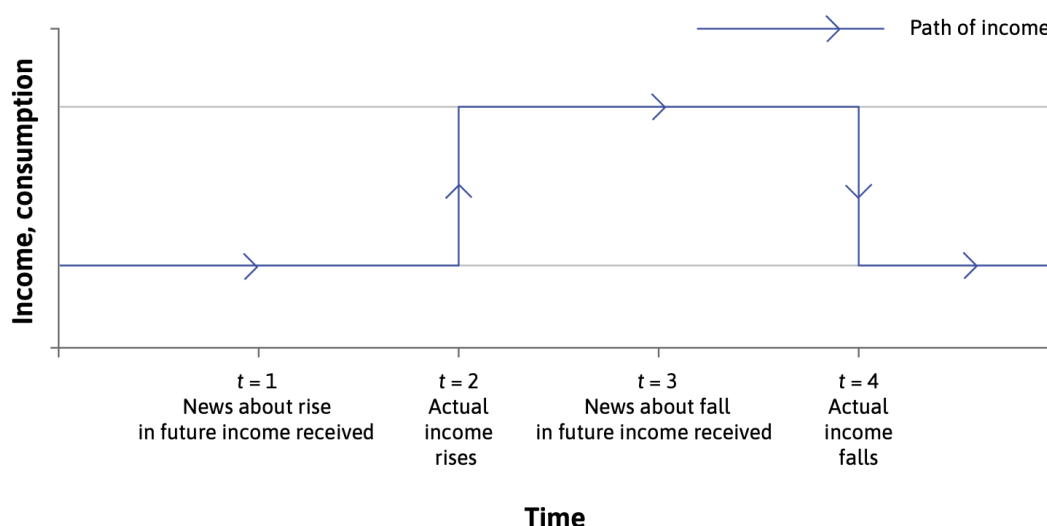
5. Which of the following statements is correct regarding measuring GDP?

- a) GDP can be measured either as the total spending on domestically produced goods and services, or the total value added in domestic production, or the sum of all incomes received from domestic production.
- b) Information about exports but not imports is necessary to calculate GDP.
- c) Government production is not included in the GDP.
- d) The value added of government production is computed using the price that public goods and services are sold at in the market.

6. Which of the following would increase GDP?

- a) A decline in imports, holding all other components of GDP constant.
- b) An increase in remittances paid to domestic residents by relatives living abroad.
- c) An increase in government spending.
- d) A decline in exports.

7. Since 2008 we have got used to the idea that the size and behaviour of the financial system can be a major source of economic instability. But can you think of ways in which the existence of a developed financial system may help to limit fluctuations in the economy?
- A developed financial system means that firms and households can always borrow.
 - A developed financial system can help households to 'smooth' their consumption over time.
 - A developed financial system facilitates government borrowing.
 - A developed financial system provides facilities for trading currencies and for lending and borrowing overseas.
8. Consider the following two alternative definitions of a recession: A period of negative output growth. A period when the output level is below the country's potential, or normal, level. Now consider a country that has been producing a lot of oil, whose oil wells have suddenly run out. This results in both its potential, and actual, output falling. Which of the following statements are correct?
- According to the first definition, the country is now unambiguously in a recession.
 - According to the second definition, the country is now unambiguously in a recession.
 - The policymaker using the first definition may underreact to the slowing output.
 - The policymaker using the second definition may overreact to the slowing output.
9. The following diagram shows the path of income for a household that receives news about an expected rise and fall in future income at the depicted times.



Assume that the household prefers to smooth out its consumption if it can. Based on this information, which of the following statements is correct?

- If the household is not credit-constrained, then it will consume the same level after $t = 1$.
- If the household is credit-constrained and has 'weakness of will', then its consumption will match precisely its income path.
- If the household is not credit-constrained but has 'weakness of will', then it will borrow at $t = 1$ and save at $t = 3$.

- d. If the household is credit-constrained but does not have ‘weakness of will’, then it will borrow at $t = 1$ and save at $t = 3$.

3. CONCEPTUAL QUESTIONS

10. Think about the health insurance system in your country. Is this an example of co-insurance or self-insurance?
11. Can you think of other examples of both co-insurance and self-insurance? In each case, consider what kinds of shocks are being insured against and how the scheme is financed.?